

# DEXTER FERNANDES

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## SUMMARY

Software engineer with 3+ years building and deploying Python-based AI and perception systems, backed by an MSc Robotics. Integrated C++17 and ROS2 perception, SLAM and navigation software on a Clearpath Husky and NVIDIA Jetson. Commercial work spans Dockerised cloud and edge services, REST APIs, PostgreSQL, Kafka, observability, automated testing and 24/7 field deployments.

## RELEVANT EXPERIENCE

### Esbaar - AI Engineer

01/2023 - 08/2023 | MUSCAT, OMAN

- Developed and maintained NVIDIA DeepStream 6.x and FFmpeg pipelines for 24/7 perception on BP drilling rigs, covering camera ingestion, inference and field diagnostics.
- Hardened deployed services with Dockerised microservices, Kafka, health checks and safe restarts, cutting live video pipeline crashes by 67% and enabling remote recovery.
- Built a watchdog and REST API exposing service health and restart controls, allowing operators and connected services to recover failed edge processes without site visits.
- Added system telemetry through structured logs, metrics, timeouts, retries and message validation, shortening root-cause analysis on fielded services.
- Introduced pytest, unittest and linting into CI for Python edge-inference modules, improving release consistency across hardware targets.
- Adapted YOLOv5 models in Python and PyTorch and owned the lifecycle from data curation through edge deployment and on-site validation, improving F1 by 15%.

### LivNSense Technologies - Lead Software Engineer, AI

05/2020 - 12/2022 | BENGALURU, INDIA

- Built asynchronous C++17 perception pipelines with OpenVINO on Intel VPUs and industrial edge computers, raising throughput from 5 to 15 FPS under real-time constraints.
- Deployed containerised ML inference microservices on Microsoft Azure with REST endpoints and queue-based processing, owning autoscaling, monitoring and cost-per-inference.
- Designed the PostgreSQL schema routing real-time safety incidents into downstream analytics and wrote the joins and CTEs supporting operational reporting.
- Wrote gtest unit tests and documented interfaces for the C++ inference path, catching regressions before release and keeping hardware deployments reproducible.
- Led four engineers delivering a two-stage detector and helmet-colour classifier for construction safety, cutting false positives by 33% within latency and power budgets.
- Worked with hardware, software and QA teams to integrate perception modules into deployed systems and convert field findings into rollout decisions.

## PROJECTS

### ROS2 SLAM and Nav2 Simulation

02/2026

- Built a robotics stack in ROS2 Jazzy and Gazebo Harmonic to evaluate LiDAR SLAM, LiDAR/RGB fusion and autonomous navigation in structured simulations.
- Developed C++ and Python ROS2 nodes, launch files and parameterised configurations, using tf2, message synchronisation and timestamp alignment to debug transform timing.
- Implemented PCL and Open3D point-cloud processing plus EKF localisation with robot\_localization, then containerised the Linux stack with Docker for reproducible builds.

### DeepStream RTSP Pipeline [github.com/Dexter-Fernandes/deepstream-rtsp-pipeline](https://github.com/Dexter-Fernandes/deepstream-rtsp-pipeline)

06/2026

- Built an end-to-end DeepStream 9.0 and GStreamer RTSP pipeline running YOLO detection and NvDCF tracking across three feeds, with metadata extraction and re-streamed output.
- Converted YOLO26n to a TensorRT FP16 engine supporting variable batch sizes on 6 GB VRAM hardware, validating accuracy and profiling p99 latency to sustain 29.7 FPS per live stream.
- Implemented a C++ and CUDA decode plugin through the TensorRT plugin API, moving post-processing into the engine for custom-layer deployment.
- Benchmarked IoU, NvDCF and NvSORT trackers on MOTA, IDF1, identity switches, fragmentation, FPS and VRAM use to guide tracker selection.

### Modern Edge CV Benchmarking [github.com/Dexter-Fernandes/ModernEdge-CV-Benchmarking](https://github.com/Dexter-Fernandes/ModernEdge-CV-Benchmarking)

03/2026

- Built a repeatable PyTorch benchmarking harness comparing YOLO, RT-DETR and other vision models on accuracy, latency and model size for edge deployment.
- Exported models to ONNX and tested OpenVINO paths, comparing mAP, inference time and model size to support runtime and hardware selection.
- Extended the harness to segmentation and depth models to assess whether broader perception workloads remained viable under the same edge constraints.

EDUCATION

- University of Bristol - MSc Robotics, Merit09/2023 - 09/2024 | BRISTOL, UK
- Dissertation: Radiological SLAM with LiDAR Odometry
- Built a real-time SLAM system in C++17 and ROS2 on a Clearpath Husky, fusing Velodyne LiDAR, RealSense depth and radiological sensing for inspection mapping.
  - Benchmarked RTAB-Map, LIO-SAM and Cartographer on matched rosbag2 sequences under one evaluation protocol, comparing mapping and localisation behaviour across configurations.
  - Configured Ceres, g2o and GTSAM solver backends, tuned ICP parameters, calibrated LiDAR extrinsics and resolved transform timing with tf2 across a PCL pipeline.
  - Ran the full ROS2 perception stack on an NVIDIA Jetson AGX Xavier within its compute and memory limits, then configured Nav2 waypoint traversal over the generated map.

- Visvesvaraya National Institute of Technology - BTech Electrical and Electronics Engineering07/2016 - 05/2020 | NAGPUR, INDIA
- Studied signals, systems, embedded hardware and control, providing the foundation for real-time AI, sensor systems and robotics software.

SKILLS

|                          |                                                                                                                                                                           |
|--------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Languages                | Python, C++17, CUDA C++, Bash, SQL                                                                                                                                        |
| Robotics                 | ROS2 (Robot Operating System 2), rclcpp, rclpy, tf2, rosbag2, Gazebo, Nav2, robot_localization, EKF, Clearpath Husky                                                      |
| SLAM and sensors         | RTAB-Map, LIO-SAM, Cartographer, LiDAR SLAM, ICP, scan matching, pose graph optimisation, PCL, Open3D, Eigen, Ceres, g2o, GTSAM, LiDAR extrinsics calibration             |
| AI and perception        | PyTorch, TensorFlow, Keras, OpenCV, scikit-image, YOLO, RT-DETR, object detection, tracking, optical flow, sensor fusion, image preprocessing                             |
| Inference and video      | TensorRT, OpenVINO, ONNX, DeepStream, GStreamer, FFmpeg, RTSP, FP16/INT8, variable batching, quantisation, pruning, NVIDIA Jetson, Intel Movidius VPU                     |
| Cloud and backend        | Microsoft Azure, Container Apps, Blob Storage, queues, REST API design, PostgreSQL, Django, Kafka, Docker, autoscaling, containerised microservices                       |
| Testing and delivery     | pytest, unittest, gtest, CI/CD, Git, CMake, benchmarking, failure-mode analysis, code reviews, Agile and Scrum                                                            |
| MLOps                    | MLflow, DVC, Airflow, Optuna, Weights & Biases, CVAT, dataset versioning, experiment tracking, automated labelling, drift tracking                                        |
| Telemetry and operations | structured logging, metrics, health checks, observability, production monitoring, timeouts, retries, message validation, multiprocessing, GPU and edge hardware profiling |

ADDITIONAL EXPERIENCE

- GrowthStage - Technical Assessor, Computer Vision and Software Engineering05/2026 - PRESENT | REMOTE
- Assess candidate depth across perception, SLAM and software deployment using hands-on robotics and computer vision knowledge.
- Taco Bell - Service Champion / Team Trainer09/2024 - PRESENT | BRISTOL, UK
- Train new staff and handle customer escalations after progressing to Team Trainer.

CERTIFICATIONS AND AWARDS

Deep Learning Specialisation, deeplearning.ai (07/2019)      Best Speaker award, ABC Toastmasters Club (01/2020)